Multi-Task Learning for Stock Price Prediction

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Abstract

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I. Introduction

Insert introduction here

II. Motivation and Objectives

Motivation

Objectives

III. Methodology

A. Data

- Companies
- aggregation choice
- time choice

B. Algorithms

LSTMS: explination

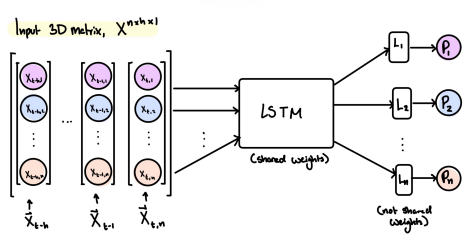

Fig. 1. Diagram of hard parameter sharing and inputs for an LSTM Multi-Tasking model. (Drawn by group).

Multi-Tasking Extra Trees (or Extremely Randomized Trees): Extra Trees is a tree ensemble method like Random Forests, but it produces more decorrelated trees. It uses the entire data set to build decision trees, rather than bootstrapping as in random forests. It also differs in its threshold selection, which randomly selects where to split the feature rather than using an optimization method. It still chooses m_{try} number of random features to choose the feature splits to compare, where $m_{try} = \lfloor \frac{p}{3} \rfloor$ and p is the number of features. There are two relevant extensions to this algorithm that make it similar to our LSTM hard parameter multi-tasking [1].

The first is introducing multi-output, which essentially means the loss function is the summation of sample error and task outputs. This effectively makes each node split consider the error of the candiate splits' effect across all tasks [1]

The second is introducing task-focused splits, or the ability to make decisions based on what the task is being predicted. To do this, a random group of m_{try} features (in this case, the lags) and S random possible subsets of tasks are selected (S$\epsilon \mathbb{Z} < T$ randomly selected). Each feature split is randomized, and the multi-output error is calculated. Then, f_k (1) is calculated for each candidate task subset or feature; this is the prediction. The error is calculated from the prediction and used to minimize, allowing the best split to be chosen [1].

$$E = mc^2 \tag{1}$$

C. Metrics and Evaluation

D. Preprocessing

Misaligned Dates: Due to different market holidays, some days in the data included both null values and recorded prices. The days with null values were removed to keep the dataset consistent. Dropping a few days of data should not obscure the stocks' growth patterns and relationships.

Logarithmic Scale: The data was scaled logarithmically. This was to ensure stable predictions by reducing effect of outliers on the model and to account for the exponential patterns in each stock's data. The predictions become more stable because most analysis techniques assume additive data, including machine learning. [2].

The scaling affect is shown in appendix A, the medium stock ACIW is on a similar scale to the large stock AAPL after logging the data.

LSTMS: Pytorch requires LSTM inputs to be 3D tensors. The data looks at stocks, with the only attribute being the stock price. Referring to 1, the matrix on the left would be an example 3D tensor input, with each vector $\vec{x}$ representing all the stocks of interest at the given t. There is no third dimension depicted, as the only attribute of each stock of interest is the given price at time t.

The times within the input are $\vec{x_{t,n}}$ and the lags of $\vec{x_{tn}}$, which represent our rolling window, since $\vec{x_{t+1}}$ should use all the previous lags for its prediction.

Extra Trees: describe script and set up of what it intakes [2]

E. Building the Models (?)

Random Forests:

LSTMS:

IV. Results

add pics and stuff

A. Hyperparameters

B. Algorithm Performances

V. Discussion

A. Findings

B. Interpretation of Results

C. Hyperparameters

D. Advantages and Disadvantages of Multi-Tasking

VI. Conclusion

References

[1] L. Schmid, A. Gerharz, A. Groll, and M. Pauly, "Tree-based ensembles for multi-output regression: Comparing multivariate approaches with separate univariate ones," *Computational Statistics & Data Analysis*, vol. 179, p. 107628, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0167947322002080

[2] "Decomposition of time series," 2025. [Online]. Available: https://medium.com/@m.capurso/decomposition-of-time-series-071ce7b7a5a0

Appendix

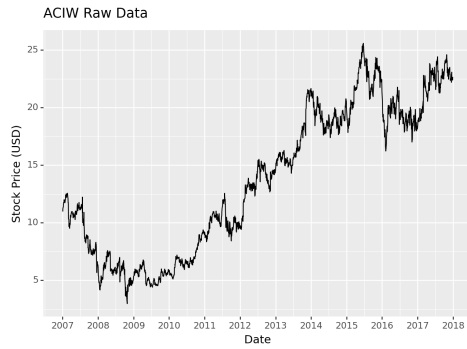

Fig. A2. Raw price data of ACIW Stock

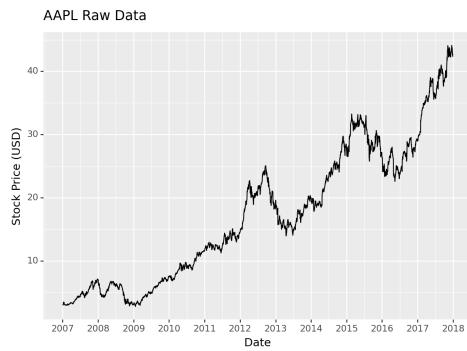

Fig. A3. Raw price data of APPL Stock

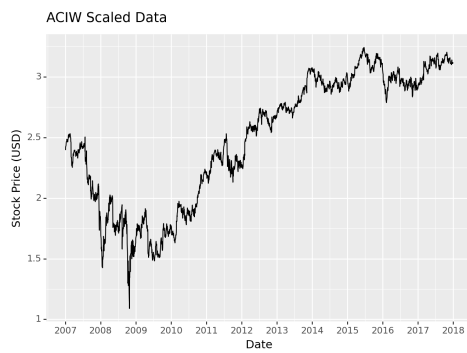

Fig. A4. Logged price data of ACIW Stock

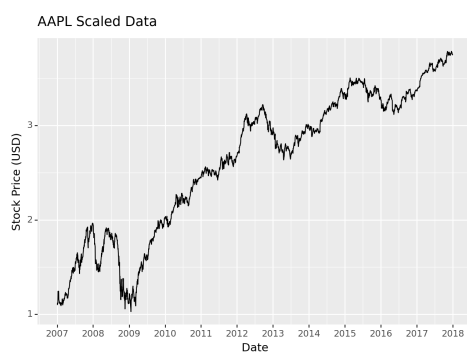

Fig. A5. Logged price data of APPL Stock